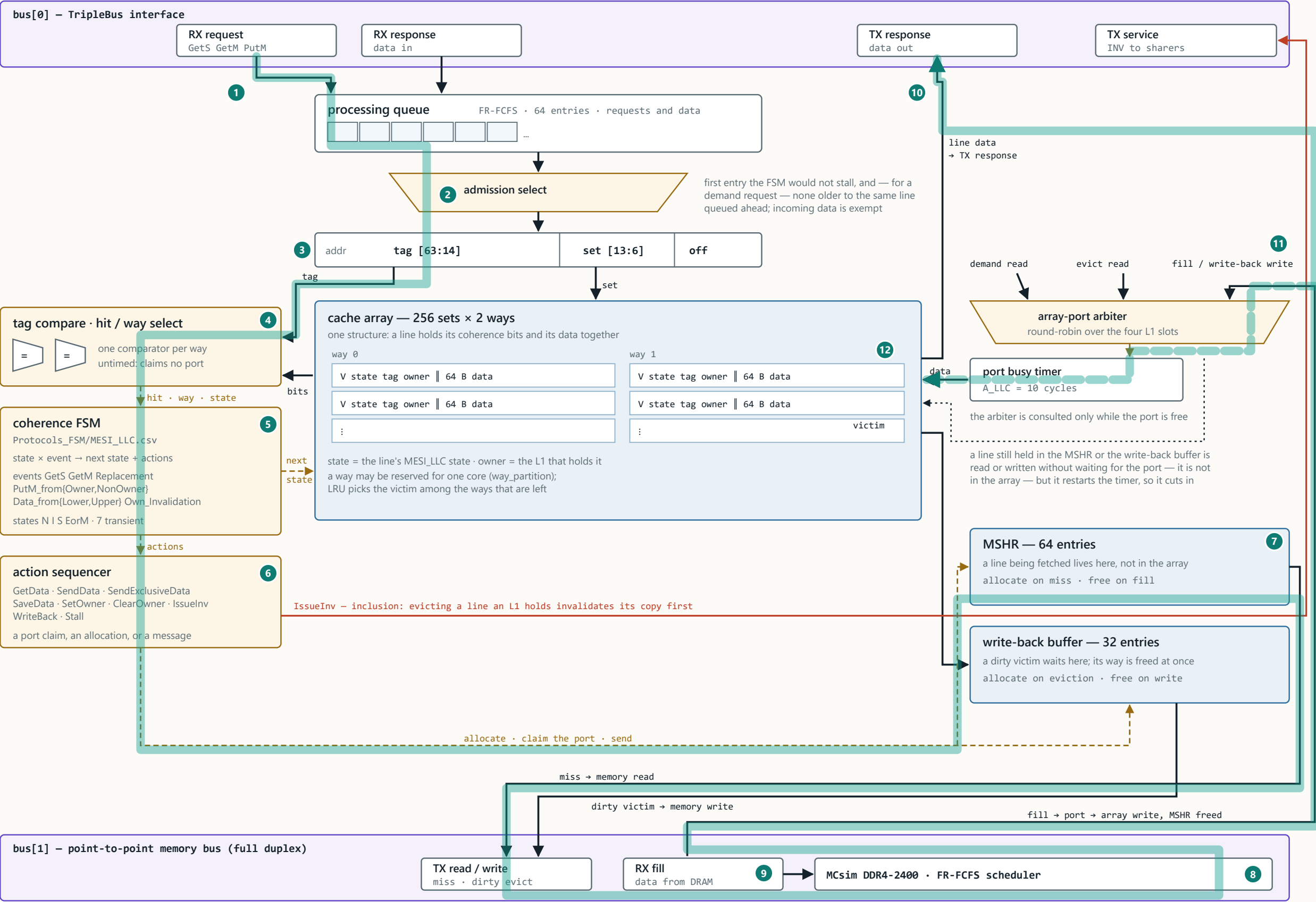


Octopus LLC datapath — the path of a read miss (free way, no eviction)

CacheController_End2End + CacheDataHandler_COTS · 32 KB, 2-way, 64 B lines · MESI_LLC coherence FSM · one timed data port

Highlighted: a GetS that misses in a set with a free way. The response leaves as soon as the fill is processed; the array write into the free way follows in parallel (dashed). Nothing is evicted, so the write-back buffer and the red inclusion path stay idle.



— data path (timed) register-resident access, bypasses the port - - - - - control (untimed) — back-invalidation (inclusion)

□ storage □ control logic □ interconnect interface the Logger and the raw trace observe: queue arrival · admission · port claim · data emitted

read miss, free way — 1 arrives · 2 admitted · 3 decoded · 4 tag miss; a free way, no victim · 5 FSM: GetData, SetOwner · 6 sequencer: allocate, send · 7 MSHR · 8 DRAM read · 9 fill on RX · 10 response out, from the fill itself · off the critical path: 11 port claim → busy timer · 12 written into the free way, MSHR freed